AP BIOLOGY TOPIC 2.8

Mechanisms of Transport: Primary vs. Secondary Active Transport & Pumps

UNIT 2: CELL STRUCTURE AND FUNCTION (10–13% EXAM WEIGHT)

ENE-2.J | NA^+/K^+ ATPASE | COTRANSPORT (SYMPORT/ANTIPORT) | H^+ PROTON MOTIVE FORCE

Credits: mitali120501@gmail.com

§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 2.8 (MECHANISMS OF TRANSPORT: PRIMARY VS. SECONDARY ACTIVE TRANSPORT & PUMPS)

CED Core Vocabulary

Two Types of Secondary Active Cotransporters — Symport (Symporter): Both the driving ion and the co-transported solute move in the SAME direction across the membrane (e.g., Na^+ and glucose both enter...

Primary Active Transport = Directly uses ATP — to build an ion gradient.

Electrogenic Effect — Because 3 positive charges leave for every 2 positive charges that enter (net export of +1 charge per cycle), it directly generates a net negative charge inside...

Step 2 — ATP hydrolyzes (ATP → ADP + P_i) , covalently phosphorylating the pump (Aspartate residue).

Antiport (Antiporter / Exchanger) — The driving ion moves downhill in one direction while the second ion/solute is driven uphill in the OPPOSITE direction (e.g., Na^+/Ca^{2+} exchanger in cardiac...

Stoichiometry per Cycle — Exports 3 Na^+ OUT of the cell and imports 2 K^+ INTO the cell at the cost of 1 ATP hydrolyzed.

4 Step Conformational Cycle — Step 1: Cytoplasmic 3 Na^+ bind high-affinity sites on the inward-facing pump.

Step 3 — Phosphorylation triggers a conformational shape change to outward-facing → releases 3 Na^+ outside and binds extracellular 2 K^+ .

§2. PRIMARY VS. SECONDARY ACTIVE TRANSPORT (COTRANSPORT)

Energy Coupling

- **Role of Integral Membrane Proteins in Active Transport:**
  - Integral carrier proteins (**pumps & cotransporters**) are highly specific for their target ions/solutes and undergo reversible **conformational shape changes** to move substances uphill against their concentration or electrochemical gradient (**[Low] → [High]**).

FEATURE	PRIMARY ACTIVE TRANSPORT	SECONDARY ACTIVE TRANSPORT (COTRANSPORT)
Direct Energy Source	**Direct ATP hydrolysis** (**ATP → ADP + P_i**); phosphate group directly phosphorylates the pump	**Potential energy stored in an electrochemical gradient** (created earlier by a primary ATP pump; no direct ATP binding)
Mechanism	Pumps ion(s) uphill (**Low → High**) to build a steep electrochemical gradient across the membrane	Driving ion (**Na^+** or **H^+**) diffuses **downhill** (**High → Low**) through cotransporter, pulling a second solute **uphill** (**Low → High**)
Classic Examples	**Na^+/K^+ ATPase pump** (animal cells), **H^+ proton pump** (plants, fungi, bacteria, lysosomes), **Ca^{2+}** ATPase (SERCA)	**Na^+ /Glucose Symporter (SGLT1)** in gut/kidney; **H^+ /Sucrose Cotransporter** in plant phloem loading

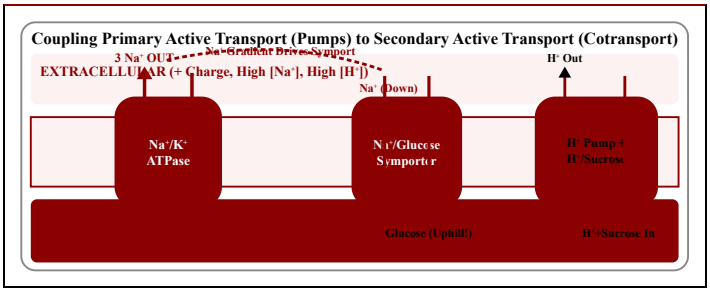
- **Two Types of Secondary Active Cotransporters:**
  - **Symport (Symporter):** Both the driving ion and the co-transported solute move in the **SAME direction** across the membrane (e.g., **Na^+** and glucose both enter intestinal epithelial cells via **SGLT1**).
  - **Antiport (Antiporter / Exchanger):** The driving ion moves downhill in one direction while the second ion/solute is driven uphill in the **OPPOSITE direction** (e.g., **Na^+/Ca^{2+}** exchanger in cardiac muscle: **Na^+** enters downhill while **Ca^{2+}** is exported uphill).

[CORE AXIOM / REMEMBER]

- **Primary Active Transport = Directly uses ATP** to build an ion gradient.
- **Secondary Active Transport (Cotransport) = Indirectly uses ATP** by harvesting the electrochemical gradient built by the primary pump! (If ATP production stops via cyanide/dinitrophenol, secondary active transport soon stops too as the gradient runs down!)

§4. NA^+/K^+ PUMP & COUPLED COTRANSPORT VISUALIZED

Primary → Secondary Coupling



§3. THE SODIUM-POTASSIUM PUMP (NA^+/K^+ ATPASE)

3 Na^+ Out / 2 K^+ In

- **Classic Primary Active Electorgenic Pump of Animal Cells:**
  - **Stoichiometry per Cycle:** Exports **3 Na^+ OUT of the cell** and imports **2 K^+ INTO the cell** at the cost of **1 ATP** hydrolyzed.
  - **Both ions move AGAINST their concentration gradients:** Maintains **high [Na^+] outside** (**~145 mM** out vs. **12 mM** in) and **high [K^+] inside** (**~140 mM** in vs. **4 mM** out).
  - **Electrogenic Effect:** Because **3** positive charges leave for every **2** positive charges that enter (net export of **+1** charge per cycle), it directly generates a **net negative charge inside the cell**!
- **4 Step Conformational Cycle:**
  - **Step 1:** Cytoplasmic **3 Na^+** bind high-affinity sites on the inward-facing pump.
  - **Step 2:** **ATP hydrolyzes (ATP → ADP + P_i)**, covalently phosphorylating the pump (Aspartate residue).
  - **Step 3:** Phosphorylation triggers a **conformational shape change** to outward-facing → releases **3 Na^+** outside and binds extracellular **2 K^+**.
  - **Step 4:** Binding of **2 K^+** triggers **dephosphorylation (release of P_i)** → pump returns to original shape and releases **2 K^+** into the cytosol.

§5. ELECTROCHEMICAL GRADIENTS, MEMBRANE POTENTIAL & H^+ PUMPS

Bioenergetics

- **Electrochemical Gradients Store Potential Energy:**
  - Combines the **concentration gradient (chemical force)** and the **membrane voltage (electrical force)**.
  - Drives three essential cellular processes: **(1) rapid ion flux in signaling**, **(2) secondary active cotransport** of nutrients, and **(3) chemiosmotic ATP synthesis**.
- **Membrane Potential ($V_m \approx -70$ mV in Animal Cells):**
  - Established by the **Na^+/K^+ ATPase pump** + open **K^+ leak channels**.
  - Essential for **action potentials (nerve impulses)** in neurons and **excitation-contraction coupling** in skeletal/cardiac muscle cells.
- **Proton Gradients (H^+ Pumps & Proton-Motive Force):**
  - **In Plants, Fungi & Bacteria:** The **Proton Pump (H^+-ATPase)** is the main electrogenic pump on the plasma membrane — uses ATP to pump **H^+** out of the cell, driving **H^+/sucrose cotransport** (phloem loading) and root mineral (**NO_3^-**) uptake.
  - **In Lysosomes & Plant Vacuoles:** V-type **H^+**-ATPase pumps **H^+** inside to maintain an **acidic lumen (pH ~4.8)** for hydrolytic enzymes.
  - **In Mitochondria & Chloroplasts (Chemiosmosis):** Electron transport chains pump **H^+** across the inner mitochondrial membrane (into the intermembrane space) or across the thylakoid membrane (into the thylakoid lumen). This steep **H^+** gradient (**proton-motive force**) flows back down through **ATP Synthase** to power **ADP + P_i → ATP**!

[EXAMPLE & BIOLOGICAL APPLICATION] Oral Rehydration Therapy (ORT) Medical Application: Cholera toxin causes fatal osmotic diarrhea by locking intestinal **Cl^-** channels open. Drinking plain water fails, but drinking a solution of **both NaCl + Glucose** activates the intestinal **Na^+/Glucose Symporter (SGLT1)** → pulls **Na^+** and glucose into intestinal cells, lowering internal **Ψs** so water rapidly follows by osmosis (saving millions of lives)!

[AP EXAM FRQ SCORING STRATEGY] Cross-Unit Link (Units 2, 3, & 4):

- **Animal cells** rely primarily on the **Na^+/K^+ pump** for plasma membrane gradients.
- **Plant, fungal, & bacterial cells** rely primarily on the **H^+ proton pump** for plasma membrane cotransport.
- **Mitochondria & Chloroplasts** use **ETC-driven H^+ gradients** to drive **ATP synthase** (facilitated diffusion of **H^+** coupled to phosphorylation!).

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 2.8 **[MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]**

Topic 2.8: Mechanisms of Transport: Primary vs. Secondary Active Transport & Pumps ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigation 4 (Diffusion & Osmosis): Dialysis tubing & potato core mass change (% ΔMass = [(Final − Initial) / Initial] × 100) across sucrose molarities (0.0 M to 1.0 M). IV: Extracellular sucrose molarity (M); DV: % change in mass; Isotonic point: x-intercept where % ΔMass = 0% (no net water movement). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Graph Rules: Total Water Potential: $\Psi = \Psi_p + \Psi_s$. Solute Potential: $\Psi_s = -iCRT$ (where i = ionization constant [1 for sucrose/glucose, 2 for NaCl], C = molarity in mol/L, R = 0.0831 L·bar/mol·K, T = K = °C + 273). Surface-Area-to-Volume Ratio (SA:V): Cube SA = $6s^2$, V = s^3 (SA:V = 6/s). Larger SA:V accelerates diffusion per unit volume. **Error Bar Rule (±2 $SE_{\bar{x}}$):** If the ±2 $SE_{\bar{x}}$ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Physiological Disruption Case Study: Cystic Fibrosis results from a deletion (ΔF508) in the CFTR chloride channel protein, preventing misfolded channels from reaching the apical plasma membrane. Impaired Cl^- efflux prevents osmotic water movement into mucus, producing thick, dehydrated mucus in lungs and pancreatic ducts. Ouabain/Digoxin inhibits the Na^+/K^+-ATPase pump, collapsing the electrochemical gradient required for secondary active cotransport.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 2.8 to prove why the claim holds.

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